

Assignment4

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```
#load and choose numerical variables and standardize them
```

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

```
library(cluster)
```

```
pharma <- read.csv("Pharmaceuticals-1.csv")
```

```
numV <- pharma %>%select(Market_Cap, Beta, PE_Ratio, ROE, ROA,Asset_Turnover, Leverage, Rev_Growth, Net
```

```
numVS <- numV %>%scale()
```

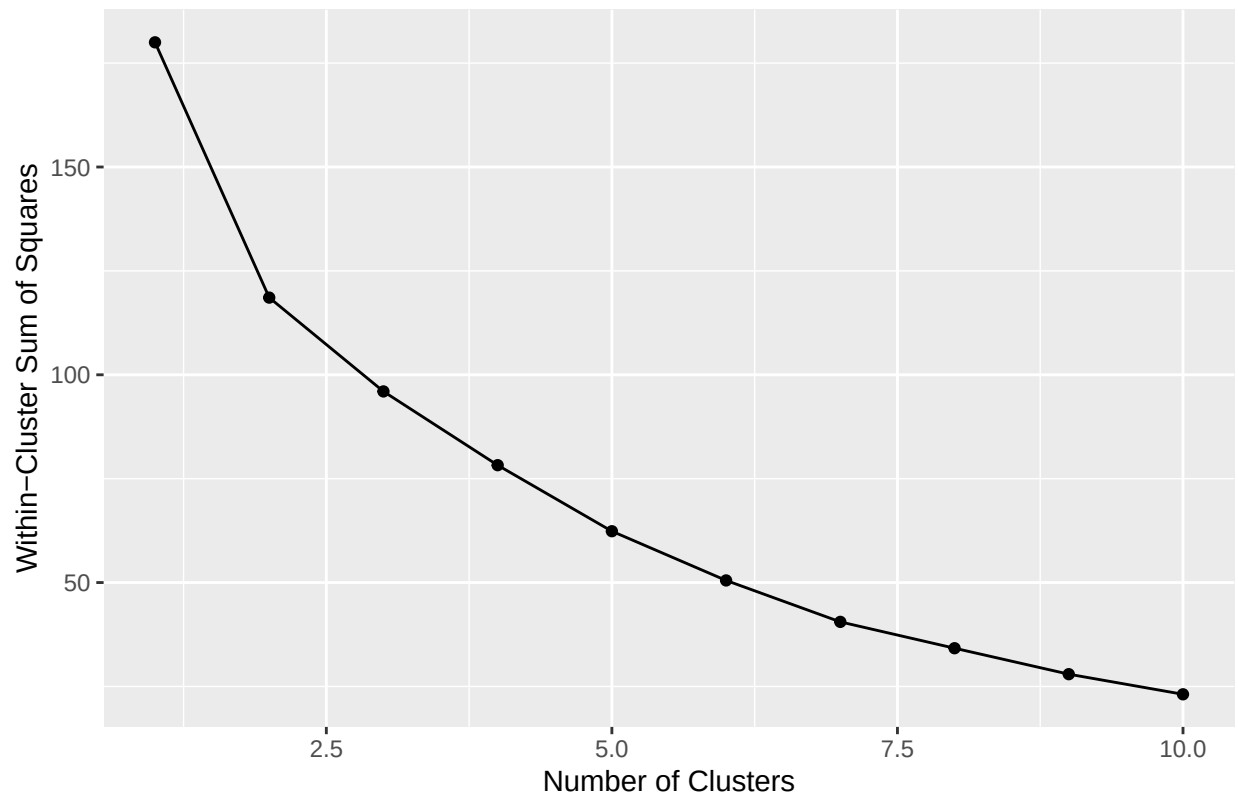
```
 #(A)use elbow method
```

```
wss <- 1:10 %>%sapply(function(k) {kmeans(numVS, centers = k, nstart = 25)$tot.withinss})
```

```
data.frame(k = 1:10, wss = wss) %>%ggplot(aes(x = k, y = wss)) +geom_line() +geom_point() +
```

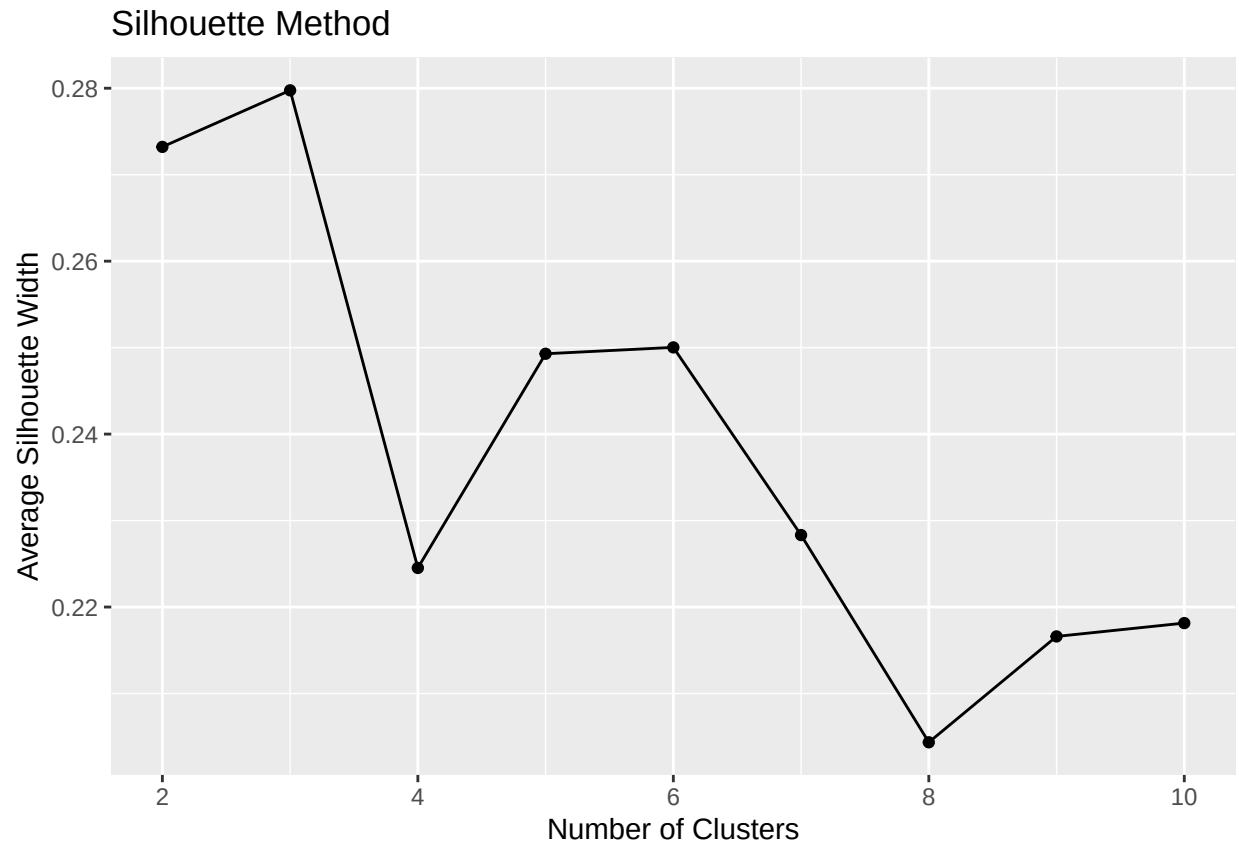
```
labs(title = "Elbow Method",x = "Number of Clusters",y = "Within-Cluster Sum of Squares")
```

Elbow Method



#use silhouette method

```
avg_sil <- 2:10 %>%apply(function(k) {km <- numVS %>%kmeans(centers = k, nstart = 25)
  silhouette(km$cluster, dist(numVS))[, 3] %>% mean()})
data.frame(k = 2:10, silhouette = avg_sil) %>%ggplot(aes(x = k, y = silhouette)) +geom_line() +
  geom_point() +labs(title = "Silhouette Method",x = "Number of Clusters",y = "Average Silhouette Width")
```



#use k=3 and add cluster to data and show which firm is in which cluster

```
set.seed(123)
km_final <- numVS %>%kmeans(centers = 3, nstart = 25)
pharma <- pharma %>%mutate(Cluster = as.factor(km_final$cluster))
pharma %>%select(Symbol, Name, Cluster)
```

##	Symbol	Name	Cluster
## 1	ABT	Abbott Laboratories	2
## 2	AGN	Allergan, Inc.	1
## 3	AHM	Amersham plc	1
## 4	AZN	AstraZeneca PLC	2
## 5	AVE	Aventis	3
## 6	BAY	Bayer AG	1
## 7	BMJ	Bristol-Myers Squibb Company	2
## 8	CHTT	Chattem, Inc	3
## 9	ELN	Elan Corporation, plc	3
## 10	LLY	Eli Lilly and Company	2
## 11	GSK	GlaxoSmithKline plc	2
## 12	IVX	IVAX Corporation	3
## 13	JNJ	Johnson & Johnson	2
## 14	MRX	Medicis Pharmaceutical Corporation	3
## 15	MRK	Merck & Co., Inc.	2
## 16	NVS	Novartis AG	2
## 17	PFE	Pfizer Inc	2
## 18	PHA	Pharmacia Corporation	1

```
## 19    SGP      Schering-Plough Corporation      2
## 20    WPI      Watson Pharmaceuticals, Inc.      3
## 21    WYE                               Wyeth      2
```

#(B) Interpret the clusters using the numerical variables

```
cluster2 <- pharma %>%group_by(Cluster) %>%summarise(across(c(Market_Cap, Beta, PE_Ratio, ROE, ROA,
  Asset_Turnover, Leverage, Rev_Growth, Net_Profit_Margin),mean, na.rm = TRUE))
```

```
## Warning: There was 1 warning in 'summarise()'.
## i In argument: 'across(...)'.
## i In group 1: 'Cluster = 1'.
## Caused by warning:
## ! The '...' argument of 'across()' is deprecated as of dplyr 1.1.0.
## Supply arguments directly to '.fns' through an anonymous function instead.
##
## # Previously
##   across(a:b, mean, na.rm = TRUE)
##
## # Now
##   across(a:b, \(x) mean(x, na.rm = TRUE))
```

```
cluster2
```

```
## # A tibble: 3 x 10
##   Cluster Market_Cap  Beta PE_Ratio  ROE  ROA Asset_Turnover Leverage
##   <fct>      <dbl> <dbl>   <dbl> <dbl> <dbl>      <dbl>      <dbl>
## 1 1          21.8  0.595    46.9  11.3  5.1         0.75      0.305
## 2 2          97.1  0.434    21.0  35.7 15.0         0.8       0.325
## 3 3           9.24 0.648    19.4  17.3  5.98        0.483     1.25
## # i 2 more variables: Rev_Growth <dbl>, Net_Profit_Margin <dbl>
```

#(C) Check pattern for variables not used in clustering

```
pharma %>%count(Cluster, Median_Recommendation)
```

```
##   Cluster Median_Recommendation n
## 1      1                      Hold 2
## 2      1          Moderate Buy 1
## 3      1          Strong Buy 1
## 4      2                      Hold 6
## 5      2          Moderate Buy 3
## 6      2          Moderate Sell 2
## 7      3                      Hold 1
## 8      3          Moderate Buy 3
## 9      3          Moderate Sell 2
```

```
pharma %>%count(Cluster, Location)
```

```
##   Cluster  Location n
## 1      1    CANADA 1
```

```
## 2      1      GERMANY 1
## 3      1      UK 1
## 4      1      US 1
## 5      2 SWITZERLAND 1
## 6      2      UK 2
## 7      2      US 8
## 8      3      FRANCE 1
## 9      3      IRELAND 1
## 10     3      US 4
```

```
pharma %>%count(Cluster, Exchange)
```

```
##   Cluster Exchange  n
## 1      1      NYSE  4
## 2      2      NYSE 11
## 3      3      AMEX  1
## 4      3    NASDAQ  1
## 5      3      NYSE  4
```

#(D) Provide an appropriate name for each cluster using any or all of the variables in the dataset.

```
clusterN <- cluster2 %>%mutate(ClusterN= case_when(Market_Cap == max(Market_Cap) ~ "Large Firms",Rev_Gr
pharma <- pharma %>%left_join(clusterN%>% select(Cluster, ClusterN), by = "Cluster")
clusterN %>%select(Cluster, ClusterN)
```

```
## # A tibble: 3 x 2
##   Cluster ClusterN
##   <fct>   <chr>
## 1 1      Other Firms
## 2 2      Large Firms
## 3 3      Growth Firms
```